

LESSON 12.2

MULTIPLYING MONOMIALS BY ALGEBRAIC EXPRESSIONS



LESSON INTRODUCTION

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

MA.8.AR.1.2 Apply Properties of Operations to multiply two linear expressions with rational coefficients.

LEARNING TARGETS

- I can apply properties of operations to multiply two linear expressions.
- I can multiply a monomial by an algebraic expression.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.1.1 Apply Properties of Operations to add and subtract linear expressions with rational coefficients

WORKING TOWARD

MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.

ESSENTIAL QUESTIONS

- What is a monomial?
- How do I multiply a monomial by an algebraic expression?

LESSON NARRATIVE

In this lesson, students will work in collaboration to identify the features of a monomial to be able to create a working definition for the term. Students will build on their learning of generating equivalent expressions using the distributive property method and area model method to be able to multiply a monomial by an algebraic expression.

COMPONENT SUMMARY

12.2.1 WARM-UP (~5 MIN.)

Explain which expression does not belong

12.2.3 COLLABORATION (10-20 MIN.)

Define a monomial

12.2.5 GUIDED PRACTICE (10-15 MIN.)

Find the product of a monomial and a linear expression

12.2.7 WRAP-UP (~5 MIN.)

Analyze student work to identify the error made

12.2.2 COLLABORATION (5 MIN.)

Apply the distributive property to rewrite expressions

12.2.4 GUIDED INSTRUCTION (5-10 MIN.)

Multiply a monomial by a linear expression

12.2.6 YOUR TURN! (10 MIN.)

Find the product of a monomial and a linear expression

RESOURCES

GLOSSARY

Key Terms:

- Monomial

Additional Terms:

- N/A

MATERIALS

- (Optional) Markers or Colored Pencils

ADDITIONAL PREPARATION

- Consider strategic pairing of students for 12.2.2 and 12.2.3 Collaborations

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may believe if two terms are in parentheses, they can combine them. Be prepared to remind students what like terms are if they forget.

THINGS TO CONSIDER

- Students do not need to define the term polynomial. The BEST glossary definition was intentionally not included in the lesson since it refers to polynomials and students are not exposed to polynomials until Algebra 1.
- If time is an issue, consider assigning 12.2.6 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

12.2.2 and 12.2.3 Collaborations allow for several opportunities for students to engage in a Think-Pair-Share routine. If scaffolding is being done for the first expressions in 12.2.2 Collaboration, students can think about how they would apply the distributive property, then share their strategy with their partner to compare. Encouraging students to start the activity with a Think-Pair-Share can break down barriers for students who did not feel confident about the prior lesson, and gives them an opportunity to share with a peer before formally beginning the activity.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.MTR.4.1 Discussion and Reflection

12.2.3 Collaboration has students discuss and reflect with multiple partners. Students work to define a monomial with one partner, then share and discuss with two other partner pairs. Students work with their original partner to reflect on the discussions with their peers to decide what revisions, if any, are needed to their definition of monomial.

DIFFERENTIATE INSTRUCTION

A Frayer Model can be utilized to allow students to graphically represent the meaning of the term monomial from 12.2.3 Collaboration. Alternate representations of vocabulary help to clarify meaning for students.

Consider strategic pairing for 12.2.2 and 12.2.3 Collaborations. For example, if a student missed the previous lesson, you would not want to pair them with another student who missed the lesson too. Based on formative assessment data from the previous lesson, decide how to group students so that struggling students can successfully continue to build their understanding of how to apply the distributive property.

Definition	Characteristics
Examples	Non-examples

12.2.1 WARM-UP: WHICH ONE DOESN'T BELONG?

TEACHER MOVES

This lesson can be lengthy depending on your student needs, so ensure no more than 5 minutes is spent on 12.2.1 Warm-Up.

Students may choose any of the expressions, as they all “don’t belong” for various reasons. Ensure student explanations are mathematically based and utilize precise mathematical terms and rules in their explanations.



12.2.1 Warm-Up: Which One Doesn't Belong?

1. Julius wrote a list of expressions using variables, numbers, and operators.

- a. Circle the expression that does not belong.

$$h(4 \cdot h)$$

$$(c + 1)(-c)$$

$$2(-3 - x)$$

$$-5(2 \cdot n)$$

- b. Explain why the expression does not belong.

Answers will vary.



12.2.2 Collaboration: Applying Properties of Operations to Multiply Linear Expressions

1. With your partner, determine who is Partner A and who is Partner B.
 - a. Apply the distributive property to re-write each expression in your column in the space provided below each given expression.

Partner A	Partner B
$4r(3 - 8r)$ $12r - 32r^2$ <i>or</i> $-32r^2 + 12r$	$2r(-16r + 6)$ $12r - 32r^2$ <i>or</i> $-32r^2 + 12r$
$-\frac{2}{3}x(6x + 9)$ $-4x^2 - 6x$ <i>or</i> $-6x - 4x^2$	$(-2x - 3)2x$ $-4x^2 - 6x$ <i>or</i> $-6x - 4x^2$
$-3c(5 - c)$ $-15c + 3c^2$ <i>or</i> $3c^2 - 15c$	$c(3c - 15)$ $-15c + 3c^2$ <i>or</i> $3c^2 - 15c$
$(2b - a)12a$ $24ab - 12a^2$ <i>or</i> $-12a^2 + 24ab$	$-6a(2a - 4b)$ $24ab - 12a^2$ <i>or</i> $-12a^2 + 24ab$
$\frac{1}{4}z(10 + 16z)$ $2.5z + 4z^2$ <i>or</i> $4z^2 + 2.5z$	$-0.5z(-8z - 5)$ $2.5z + 4z^2$ <i>or</i> $4z^2 + 2.5z$

- b. Compare your work with your partner. The expressions in each row are equivalent. If your expressions are not equivalent, work together to find and correct any error(s).

12.2.2 COLLABORATION: APPLYING PROPERTIES OF OPERATIONS TO MULTIPLY LINEAR EXPRESSIONS

DIFFERENTIATE INSTRUCTION

Think-Pair-Share

For students who need additional support on applying the distributive property, consider having students utilize a Think-Pair-Share for the first two expressions. Have students think about how they would apply the distributive property, then pair and share with their partner. If students differ in their approaches, encourage them to determine if both methods result in an equivalent answer. Then allow students to finish the activity as written.

12.2.3 COLLABORATION: DEFINING MONOMIALS

TEACHER MOVES

The BEST glossary definition for monomial was intentionally left out of the lesson because the definition references a polynomial, which students will not be introduced to until Algebra 1. For this lesson, students only need to understand what a monomial is, making no reference to polynomials.

DIFFERENTIATE INSTRUCTION

A Frayer Model can be utilized to allow students to graphically represent the meaning of a monomial. Allow students to revisit their definitions, along with the examples and non-examples throughout the lesson.

TEACHER MOVES

This is a discussion opportunity. Listen as students discuss their thinking with other students to determine student thinking to highlight, or misconceptions to address. Allow for students to make mistakes and resolve the differences in their definitions with their peers.

TEACHER MOVES

Before moving on, ensure students are able to answer question 2 correctly. Make sure to address whole-group any misconceptions or clarifications that are needed as determined by observations from 12.2.3 Collaboration.



12.2.3 Collaboration: Defining Monomials

1. The charts below show several examples and non-examples of monomials.

Monomials	Not Monomials
r^5 5 x	$2.7r^{-3}$ $3x + 2y$
$\frac{3}{4}x^2y$ $80z^{15}$	$\frac{x^2}{5y}$ $a^2 + b^2 + 10$

- a. Create a list of identifying features monomials can have versus features they cannot have.

Monomials Can Have ...	Monomials Cannot Have ...
<i>Answers will vary. Variables. Constants. Positive exponents.</i>	<i>Answers will vary. Negative exponents. Operators. Variable in denominator.</i>

- b. Compare your lists with your partner. Add any additional features you found after your discussion.
- c. Work with your partner to write a definition of a monomial.
Answers will vary. A monomial can be a number, variable, or product of a number and a variable, and any exponents are positive.

- d. Ask two other pairs of partners about the definition they wrote. Listen to their definition and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of the Definition of a Monomial	Initials
<i>Answers will vary.</i>	

- e. Review the definitions with your partner and make any necessary adjustments to your own definition.
2. Work with your partner to complete the statement. Select all options that apply.

A monomial is a

- ☐ single number
- ☐ single variable
- ☐ sum of numbers and variables
- ☐ product of numbers and variables

with

- ☐ whole number
- ☐ integer
- ☐ rational

exponents.



12.2.4 Guided Instruction: Multiplying a Monomial by a Linear Expression

Monomials can also be multiplied by linear expressions by applying properties of operations and laws of exponents.

1. Consider the monomial $-\frac{2}{5}m^2k$ and the linear expression $(4m - 7)$.
- a. Determine the product of $-\frac{2}{5}m^2k$ and $(4m - 7)$ using both methods in the table.

Method 1: Use an Area Model	Method 2: Apply the Distributive Property
$-\frac{2}{5}m^2k$ $4m$ $-\frac{8}{5}m^3k$ -7 $\frac{14}{5}m^2k$	$-\frac{2}{5}m^2k(4m - 7)$ $-\frac{8}{5}m^3k + \frac{14}{5}m^2k$

- b. Write another expression equivalent to the product.
- $\frac{14}{5}m^2k - \frac{8}{5}m^3k$

12.2.4 GUIDED INSTRUCTION: MULTIPLYING A MONOMIAL BY A LINEAR EXPRESSION

DIFFERENTIATE INSTRUCTION

Utilizing an area model to represent the expression provides a visual entry point for students as they determine equivalent expressions. Ensure students are making the connection between the area model and distributive property to solidify understanding. If necessary, refer to examples from Lesson 1.

12.2.5 GUIDED PRACTICE: MULTIPLYING A MONOMIAL BY A LINEAR EXPRESSION

DIFFERENTIATE INSTRUCTION

Allow informal observation throughout the first three activities to dictate how “guided” 12.2.5 Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

TEACHER MOVES

Students may incorrectly combine unlike terms within the parenthesis first. For example, they may state that $(-9c - 3.16) = -12.16c$. Monitor for misconceptions or errors to determine what remediation is needed (and with whom).

DIFFERENTIATE INSTRUCTION

12.2.5 Guided Practice is an opportunity for students to have voice and choice in the method they choose. If time permits, or as you are monitoring the room, ask students why they prefer the method they chose? Take the time to observe if students used both methods, or primarily stuck with one method.



12.2.5 Guided Practice: Multiplying a Monomial by a Linear Expression

Find each product using your preferred method.

$$1. -2.4c^5(-9c - 3.16)$$

$$21.6c^6 + 7.584c^5$$

$$2. \frac{3}{8}xy^4\left(\frac{1}{2}y - \frac{2}{3}\right)$$

$$\frac{3}{16}xy^5 - \frac{1}{4}xy^4$$

$$3. 4.79p^3q(0.3 + 2.1r)$$

$$1.437p^3q + 10.059p^3qr$$

$$4. (5a - 7.8)\left(-\frac{5}{2}a^4b^2\right)$$

$$-12.5a^5b^2 + 19.5a^4b^2$$

**12.2.6 Your Turn!**

Find each product using your preferred method.

- $-4y^4z\left(\frac{3}{2} - 8z\right)$
 $-6y^4z + 32y^4z^2$
- $(12.7f + 9.2)(4.01f^2g^3)$
 $50.927f^3g^3 + 36.892f^2g^3$
- $\frac{4}{5}m^4n^2\left(\frac{7}{2}m - 10\right)$
 $\frac{14}{5}m^5n^2 - 8m^4n^2$
- $(2.005 - x) \cdot (x^5yz^3)$
 $2.005x^5yz^3 - x^6yz^3$

12.2.6 YOUR TURN!**DIFFERENTIATE INSTRUCTION**

Determine from observing 12.2.5 Guided Practice if students are struggling to multiply fractions and decimals. Encourage students to talk with their partner for any strategies so that it isn't a barrier in students being successful with content. Using markers or colored pencils as students distribute, or when using the area model, can provide organization and clarity for visual learners.

**12.2.7 Wrap-Up: Error Analysis**

- Elijah multiplied the monomial $-\frac{3}{4}x^2y$ and the linear expression $\frac{2}{5}x - \frac{8}{3}$. His resulting expression is shown.

$$-\frac{3}{10}x^3y - \frac{8}{3}$$

Write a note to Elijah explaining what he did correctly and what error(s) he made when finding the product of $-\frac{3}{4}x^2y$ and $\left(\frac{2}{5}x - \frac{8}{3}\right)$.

Answers will vary.

*Dear Elijah,
It appears you have correctly distributed $-\frac{3}{4}x^2y$ to the first term and wrote the product $-\frac{3}{10}x^3y$. Good job, friend! But I think you may have forgotten to distribute $-\frac{3}{4}x^2y$ to $-\frac{8}{3}$.*

12.2.7 WRAP-UP: ERROR ANALYSIS**DIFFERENTIATE INSTRUCTION**

Consider allowing students to record an explanation or explain to their partner orally instead of the written note using video technology.

QUESTIONING

Have students create their own problems to exchange with a partner to solve. Students can then discuss their answers, the method(s) used to solve, and determine if any errors were made.